

IBD Patient face higher Incidence of Anal Dysplasia and Cancer, Hightened by HPV, Perianal Disease

Introduction: Patients with inflammatory bowel disease (IBD) have increased rates of anal cancer. We evaluated whether high-risk HPV-positive results from anal sampling or perianal disease was associated with higher anal cancer or dysplasia prevalence in IBD.

Methods: We performed a retrospective analysis of Epic Cosmos, an anonymized database of 305,065,760 patients. Patients with active diagnoses from 6/1990 to 6/2026 were stratified as Crohn's disease, ulcerative colitis, or non-IBD and compared for anal cancer/dysplasia, anal HR-HPV+ DNA, and perianal disease.

Anal HR-HPV+ status, anal cancer/dysplasia, and perianal disease were defined by ICD-10 codes. Anal cancer/dysplasia included HGSIL of anus, anal intraepithelial neoplasia III, or dysplasia of anus. Perianal disease included perianal fistula, abscess, or fissure. Chi-square testing used $p < 0.05$. Although the source population included approximately 2.4 million IBD patients, testing denominators, negative results, screening frequency, and HPV vaccination status were unavailable; findings may reflect screening, surveillance, or coding differences.

Results:

Patient Type	Anal Cancer or Dysplasia (total)	Anal Cancer or Dysplasia 5 years after Perianal Disease diagnosis	Anal cancer or dysplasia 5 years after HR-HPV diagnosis	Positive High-Risk HPV diagnosis in anal area (total)
Non-IBD (n=305,065,760)	128,173 (0.0420%)	10,195 (0.0033%)	9,404 (0.0031%)	22,867 (0.0075%)
Crohn's Disease (n=1,206,873)	2,535 (0.2100%)	575 (0.0476%)	97 (0.0080%)	197 (0.0163%)
Ulcerative Colitis (n=1,191,972)	3,227 (0.2707%)	488 (0.0409%)	149 (0.0125%)	319 (0.0268%)

All row-level Pearson chi-square tests were $p < 0.001$.

Compared with non-IBD patients, anal cancer/dysplasia was 5.0-fold higher in Crohn's disease and 6.4-fold higher in ulcerative colitis. Anal HR-HPV positivity was 2.2-fold higher in Crohn's disease and 3.6-fold higher in ulcerative colitis. Anal cancer/dysplasia within 5 years after anal HR-HPV diagnosis was also higher in IBD, with a 2.6-fold increase in Crohn's disease and 4.1-fold increase in ulcerative colitis. The largest relative increase followed perianal disease diagnosis, with anal cancer/dysplasia within 5 years 14.3-fold higher in Crohn's disease and 12.3-fold higher in ulcerative colitis.

Unadjusted odds ratios versus non-IBD were similar: for anal cancer/dysplasia, OR 5.01 in Crohn's disease and 6.46 in ulcerative colitis; for anal HR-HPV positivity, OR 2.18 and 3.57; for anal cancer/dysplasia within 5 years after anal HR-HPV diagnosis, OR 2.61 and 4.06; and for anal cancer/dysplasia within 5 years after perianal disease diagnosis, OR 14.26 and 12.26.

Conclusion: Patients with IBD had higher observed frequencies of anal cancer/dysplasia and anal HR-HPV positivity than non-IBD patients. The strongest association was seen after perianal disease diagnosis, particularly in Crohn's disease. These findings support further study of targeted anal cancer screening strategies in IBD patients with anal HR-HPV positivity or prior perianal disease.